GENERAL PROJECTIONS: THE EXCESS OF Y CAP R_P^{-1}(R_P(Z)) OVER Y CAP Z IS BOUNDED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 0B2L: sources/stacks-chow/tags.txt, source lines 13934–13934. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let V be a vector space of dimension $n+1$. Let $Y, Z \subset \mathbf{P}(V)$ be closed subvarieties. There is a nonempty Zariski open $U \subset \mathbf{P}(V)$ such that for all closed points $p \in U$ we have*

$$Y \cap r_p^{-1}(r_p(Z)) = (Y \cap Z) \cup E$$

with $E \subset Y$ closed and $\dim(E) \leq \dim(Y) + \dim(Z) + 1 - n$.

Proof. Set $Y' = Y \setminus Y \cap Z$. Let $y \in Y'$, $z \in Z$ be closed points with $r_p(y) = r_p(z)$. Then p is on the line $\overline{yz}$ passing through y and z. Consider the finite type scheme

$$T = \{(y, z, p) \mid y \in Y', z \in Z, p \in \overline{yz}\}$$

and the morphism $T \to \mathbf{P}(V)$ given by $(y, z, p) \mapsto p$. Observe that T is irreducible and that $\dim(T) = \dim(Y) + \dim(Z) + 1$. Hence the general fibre of $T \to \mathbf{P}(V)$ has dimension at most $\dim(Y) + \dim(Z) + 1 - n$, more precisely, there exists a nonempty open $U \subset \mathbf{P}(V) \setminus (Y \cup Z)$ over which the fibre has dimension at most $\dim(Y) + \dim(Z) + 1 - n$ (Varieties, Lemma 0B2L). Let $p \in U$ be a closed point and let $F \subset T$ be the fibre of $T \to \mathbf{P}(V)$ over p. Then

$$(Y \cap r_p^{-1}(r_p(Z))) \setminus (Y \cap Z)$$

is the image of $F \to Y$, $(y, z, p) \mapsto y$. Again by Varieties, Lemma 0B2L the closure of the image of $F \to Y$ has dimension at most $\dim(Y) + \dim(Z) + 1 - n$. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.